Abundance Matrix Correlation Analysis Network Based on Hierarchical Multihead Self-Cross-Hybrid Attention for Hyperspectral Change Detection

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***Abstract*— Hyperspectral image (HSI) change detection is a technique for detecting the changes between the multitemporal HSIs of the same scene. Many existing change detection methods have achieved good results, but there still exist problems as follows: 1) mixed pixels exist in HSI due to the low spatial resolution of hyperspectral sensor and other external interference and 2) many existing deep learning-based networks cannot make full use of the correlation difference information between the bitemporal images. These problems are not conducive to further improving the accuracy of change detection. In this article, we propose an abundance matrix correlation analysis network based on hierarchical multihead self-cross-hybrid attention (AMCAN-HMSchA) for HSI change detection, which hierarchically highlights the correlation difference information at the subpixel level to detect the subtle changes. The endmember sharing-based abundance matrix learning module (AMLM) maps the changed information between bitemporal HSIs to the corresponding abundance matrices. The hierarchical MSchA extracts the enhanced difference features by constantly comparing the self-correlation with cross correlation between the abundance matrices of the HSIs. Then, the difference features are concatenated and fed into the fully connected layers to obtain the change map. Experiments on three widely used datasets show that the proposed method has superior performance compared with other state-of-the-art methods.**

***Index Terms*— Abundance matrix, change detection, correlation difference information, hyperspectral image (HSI).**

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I. Introduction

HYPERSPECTRAL image (HSI) is a 3-D data cube, which contains abundant spectral information and 2-D spatial information [1], [2]. Compared with multispectral images, HSI covers a wider range of wavelengths, which can obtain finer information of the ground objects features [3], [4]. Therefore, HSI is widely used in various remote sensing image processing technologies [5], [6], [7]. HSI change detection is the technique of detecting the changes between temporal HSIs acquired from the same scene at different times, which has been applied in many fields, such as mineral exploration [8], coastal wetlands monitoring [9], and land-cover monitoring [10].

Over the past decades, many HSI change detection approaches have been proposed with good results. The traditional algorithms can be divided into four categories as follows: 1) image algebra-based methods; 2) transformation-based methods; 3) classification-based methods; and 4) other methods.

The theory of image algebra methods is relatively simple, straightforward, and easy to understand, which usually obtains the change detection result through simple algebraic operations, such as image difference, image ratio, and change vector analysis (CVA) [11]. The CVA obtains the change vector of bitemporal images by difference calculation. The Euclidean distance of change vector represents change intensity and direction represents change content. However, it is difficult to determine the change threshold in these methods.

As for the transformation-based methods, the image is mapped to a new feature space, where the changed features are enlarged, and the unchanged features are suppressed to reduce data redundancy [12]. A principal component analysis (PCA) [13], iteratively reweighed multivariate alteration detection (IR-MAD) [14], and slow feature analysis (SFA) [15] are the most classical algorithms of transformation-based methods. IR-MAD is the extension of MAD [16], which can automatically obtain the change threshold and complete the change detection task through an iterative operation. SFA extracts invariant features from multitemporal remote sensing images and obtains the final change map through distance calculation and threshold segmentation. Nevertheless, the spectral

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information is inevitably destroyed during the transformation process.

Classification-based methods are aimed at assigning a unique class label to each pixel in the HSI, which can be further divided into postclassification comparison and compound classification methods [17]. Though the classification-based methods are able to provide information about the changed classes, the shortcomings are also obvious. The change detection result is affected by the cumulative error of classification, so the classification result has an important influence on the performance of change detection.

In addition to the above three categories, there are a series of other excellent thoughts applied to change detection that have also achieved good results. Zhan et al. [18] used tensor regression to detect changes of the fused multisource images. Li et al. [19] combined a superpixel segmentation technique with clustering to capture more subtle changes. Moreover, the spectral unmixing technique has been applied to HSI change detection. Peng et al. [20] proposed three self-paced learning-based NMF (SpNMF) unmixing methods, which are robust to noise and outliers. Peng et al. [21] discussed the application of low-rank and sparse representation in spectral unmixing.

In recent years, deep learning has been proven to have powerful feature learning and data fitting capabilities [22], [23], [24], [25] and has been successfully applied to change detection tasks. Zhan et al. [26] introduced a spectral–spatial convolutional neural network with Siamese architecture (SSCNN-S) model, in which the spectral and spatial features are well combined to improve the accuracy of change detection. Wang et al. [27] proposed a Siamese network with a spectral–spatial-wise attention mechanism (SSA-SiamNet), which can emphasize useful channel and location information to obtain refined spectral spatial features for change detection. Mou et al. [28] combined CNN and recurrent neural network (RNN) into an end-to-end recurrent CNN (ReCNN), where the spectral–spatial–temporal features are fully utilized to achieve superior change detection performance. Lin et al. [29] proposed bilinear convolutional neural networks (BCNNs), which is an end-to-end network that takes full advantage of the relations between the input bitemporal images. Song et al. [30] proposed a recurrent 3-D fully convolutional network (Re3FCN), which combines 3-D convolution with convolutional long short-term memory (ConvLSTM) to extract spectral–spatial–temporal features for HSI change detection. Yang et al. [31] proposed a deep multiscale pyramid network enhanced with spatial–spectral residual attention (DMPs^{2}raN), which can fully extract the multilevel and multiscale spatial–spectral features for detecting the complex changes. Chen et al. [32] proposed dual attentive fully convolutional Siamese network (DASNet), which captures a wide range of distinguishable feature representations through a dual attention to improve the recognition capability. Chen et al. [33] proposed a bitemporal image Transformer (BIT) for remote sensing image change detection, which models the spatial–temporal context information by Transformer.

Since the low spatial resolution of hyperspectral sensors and other external interference, mixed pixels exist in HSI. The single pixel contains several ground object categories, which

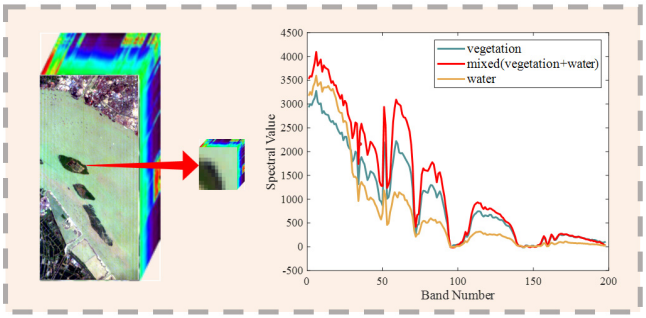


Fig. 1. Illustration of the mixed pixel in HSI.

leads to spectral aliasing. This phenomenon is particularly obvious at the boundary of ground objects. As shown in Fig. 1, the mixed pixel at the boundary of central bar contains the spectral information of vegetation and water, which does not completely belong to a certain class of ground object. This will lead to high missed and false detection rate for change detection task. In recent years, the spectral unmixing technique has been introduced in deep learning-based HSI change detection. Wang et al. [34] proposed a general end-to-end 2-D convolutional neural network (GETNET), which introduces mixed-affinity matrix that integrates subpixel representation to improve the performance of change detection. Chang et al. [9] proposed a subpixel change detection method based on collaborative coupled unmixing (SCDUM), which uses a different spatial scale nonnegative matrix factorization (NMF) for spectral unmixing of the segmented images to investigate the subtle changes. However, most of the existing deep learning methods based on spectral unmixing adopted a dual branch structure to extract the corresponding features of bitemporal images, respectively, for change detection or directly processed the difference of the abundance matrices to generate the change map. These methods may not make full use of the correlation information between bitemporal HSIs, which is not conducive to further improving the detection performance.

In this article, we propose an abundance matrix correlation analysis network based on hierarchical multihead self-cross-hybrid attention (AMCAN-HMSchA) for HSI change detection, which hierarchically highlights the correlation difference information at the subpixel level to distinguish the blurred changes. First, an abundance matrix learning module (AMLM) on the basis of endmember sharing is presented, in which the changed information of bitemporal HSIs can be mapped to the corresponding abundance matrices to solve the problem of mixed pixels. Second, we propose a hierarchical MSchA, which skillfully utilizes the correlation difference information between the abundance matrices to extract the difference feature and accurately distinguish the changed and unchanged information. The main contributions of this method are shown as follows.

1) We propose an AMCAN-HMSchA, which hierarchically enhances the correlation difference information at the subpixel level to distinguish the blurred changes between the bitemporal HSIs.
2) An endmember sharing-based AMLM is proposed, which maps the difference between bitemporal HSIs to

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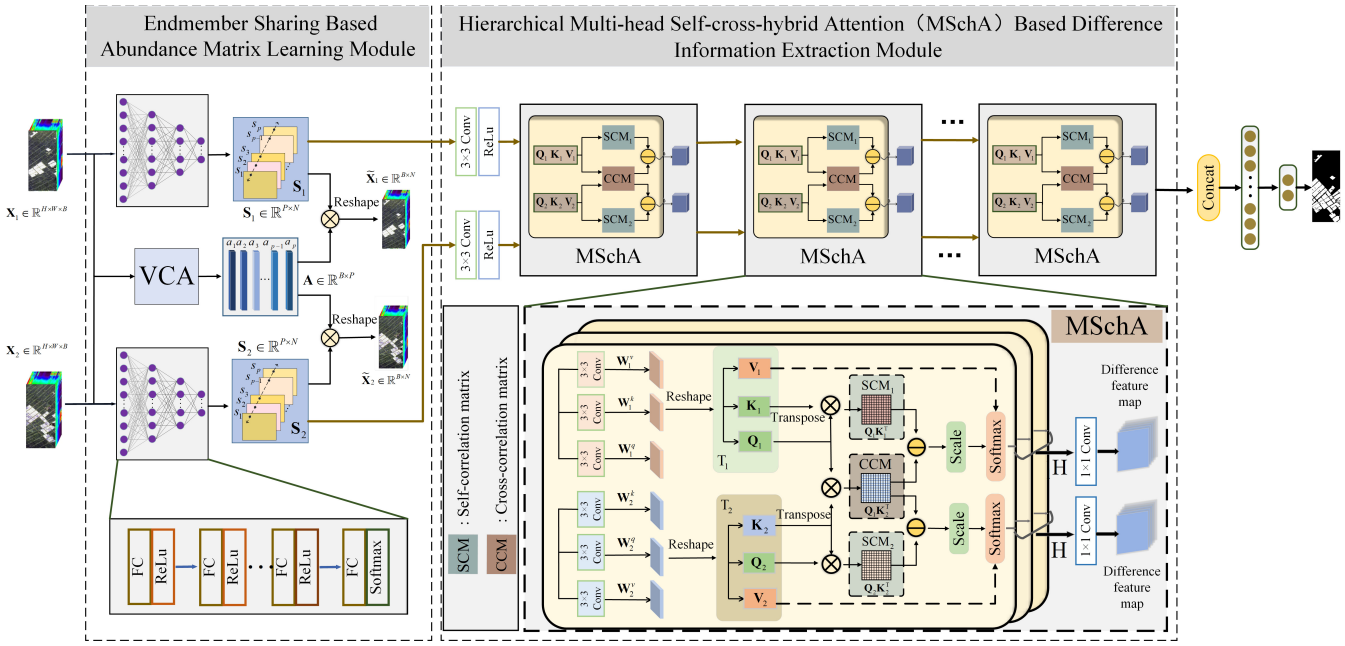


Fig. 2. Architecture of the proposed AMCAN-HMSchA for HSI change detection.

the corresponding abundance matrices for the subpixel level change detection.

3) We propose a hierarchical MSchA, in which the image features are constantly reweighted by the correlation difference between bitemporal images, so that the unchanged information can be gradually suppressed, and the changed information can be retained.

The rest of this article is arranged as follows. Section II illustrates the proposed method in detail. The performance of experiments on different datasets is represented in Section III. Finally, this article is concluded in Section IV.

II. Proposed Method

The schematic of the proposed AMCAN-HMSchA for HSI change detection is depicted in Fig. 2. The whole network consists of two modules, which are the endmember sharing-based AMLM and the hierarchical MSchA-based difference information extraction module (DIEM). Based on the spectral unmixing theory, the AMLM learns the abundance matrices of the bitemporal images under the premise of endmember sharing. Then, the DIEM extracts difference information from the learned abundance matrices through the hierarchical MSchA, in which the unchanged information is suppressed, and the changed information is highlighted. The proposed method will be described in detail below.

A. Endmember Sharing-Based AMLM

According to spectral unmixing theory, HSI can be decomposed into an endmember matrix and the corresponding abundance matrix. As shown in Fig. 3, we consider that the HSIs captured on the same scene at different times share the endmember matrix. When the endmembers are consistent, the changed and unchanged information between the two images will be synchronously mapped to the corresponding abundance matrices. In this way, the change state in the case of being mixed pixels can also be considered. Then, we analyze the correlation between the two abundance matrices to obtain the difference information, so as to complete the change detection.

The linear mixture model of HSI is shown in Fig. 4, in which the HSI can be represented as a linear combination of an endmember matrix with an abundance matrix. The mixture model of HSI is represented in matrix form as follows [35]:

$$\begin{cases} \mathbf{X} = \mathbf{AS} + \mathbf{E} \\ \text{s.t. } \mathbf{A} \geq 0, \mathbf{S} \geq 0, \quad 1_P^{\mathrm{T}}\mathbf{S} = 1_N^{\mathrm{T}} \end{cases} \tag{1}$$

where $\mathbf{X} = [\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_N] \in \mathbb{R}^{B\times N}$ represents the HSI, B is the number of spectral bands, N is the number of pixels, $\mathbf{A} = [\boldsymbol{m}_1, \boldsymbol{m}_2, \ldots, \boldsymbol{m}_P] \in \mathbb{R}^{B\times P}$ represents the endmember matrix, $\mathbf{S} = [\boldsymbol{s}_1, \boldsymbol{s}_2, \ldots, \boldsymbol{s}_N]^{\mathrm{T}} \in \mathbb{R}^{P\times N}$ is the abundance matrix, P is the number of endmembers, $\mathbf{E} = [\boldsymbol{e}_1, \boldsymbol{e}_2, \ldots, \boldsymbol{e}_N] \in \mathbb{R}^{B\times N}$ represents addictive noise, $1_P^{\mathrm{T}}\mathbf{S} = 1_N^{\mathrm{T}}$ is the sum-to-one constraint on vector of the abundance matrix, and 1_p is the $P \times 1$ vector of ones. The constraint conditions will promote the sparsity of abundance, which well reflects the sparse contribution of endmembers to the spectrum of each pixel and reduces the computation complexity.

In this article, the widely used vertex component analysis (VCA) [36] algorithm is adopted to extract the shared endmember matrix of the bitemporal HSIs. VCA is a kind of unsupervised endmember extraction algorithm based on pure pixel assumption, which is mainly based on two geometrical facts: the endmembers are the vertices of a simplex, and the affine transformation of a simplex is also a simplex. The input bitemporal HSIs is denoted as $\mathbf{X}_1 \in \mathbb{R}^{H\times W\times B}$ and $\mathbf{X}_2 \in \mathbb{R}^{H\times W\times B}$, which are acquired at the same geographical area over the different times, where H and W represent the height and width of the image, respectively, and $N = H \times W$

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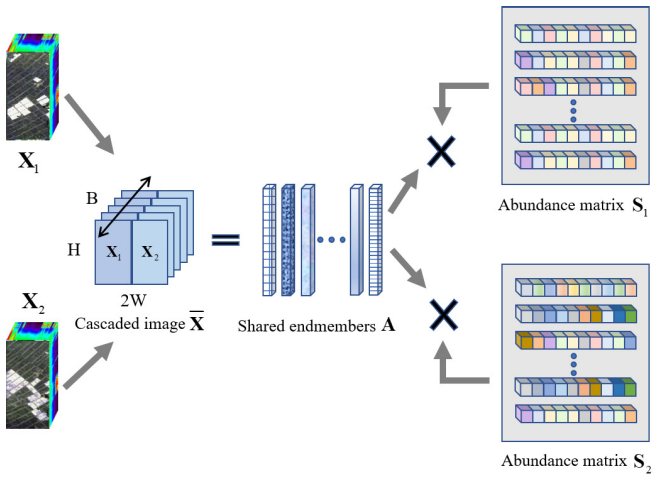


Fig. 3. Illustration of endmember sharing-based spectral unmixing.

is the number of image pixels. As shown in Fig. 3, the two input HSIs are cascaded as $\overline{\mathbf{X}} \in \mathbb{R}^{H\times 2W\times B}$ in spatial dimension to extract the shared endmembers. VCA considers that all mixed pixels of $\overline{\mathbf{X}}$ belong to a convex cone. The cone is projected onto an appropriate hyperplane, which should ensure that no observed vectors are orthogonal to it. According to the geometrical theory, the obtained projection is also a simplex. The vertices of the projection simplex are considered as endmembers. The algorithm iteratively performs this projection in the direction perpendicular to the subspace spanned by the determined endmember, and the extreme value of each iterative projection is the new endmember. The iteration process continues until the number of endmembers is exhausted, and at the same time, the complete endmember matrix $\mathbf{A}$ can be obtained.

Since the deep neural network possesses the powerful learning and data fitting capabilities, we design an abundance matrix learning block using the shared endmembers extracted above, which is mainly composed of L fully connected layers. In order to meet the constraint conditions of (1), we present to use the softmax activation function to constrain the element values of the vector in the abundance matrix between 0 and 1, and the sum of elements to 1, which correspond to the nonnegative constraint and the sum-to-one constraint, respectively.

For convenience of understanding, the abundance matrix $\mathbf{S}_i (i = 1, 2)$ can be expressed as the following:

$$\mathbf{S}_i = \sigma(g^L(\cdots g^l(\cdots g^1(\mathbf{X}_i)))), \quad i = 1, 2 \tag{2}$$

where $g^l(\cdot)$ is the function of the lth layer that contains fully connected layer and rectified linear unit (ReLU) activation, and $\sigma(\cdot)$ is the softmax activation function. In this article, L is set to 4, the dimension of the input layer is consistent with the number of image channels, and the output layer dimension is the number of endmembers shared by bitemporal HSIs.

To obtain the optimal abundance matrix $\mathbf{S}_i (i = 1, 2)$, we use the shared endmembers to reconstruct the bitemporal images separately. Specifically, the optimal abundance matrix can be obtained only when the difference between the reconstructed image $\widetilde{\mathbf{X}}_i (i = 1, 2)$ and the original image $\mathbf{X}_i (i = 1, 2)$ is minimum. Experiments has proved that L_1 norm has strong robustness owing to the insensitive to outliers [37]. Therefore,

Algorithm 1 Algorithm Flow of AMLM

Input: The number of endmembers P; The input images $\mathbf{X}_1$ and $\mathbf{X}_2$; The projection vector $\boldsymbol{w} = [1, 0, \cdots, 0]^{\mathrm{T}}$; The shared endmember matrix $\mathbf{A} = []$; The counter $j = 0$; $epoch_{\max} = 500$.

Output: The shared endmember matrix $\mathbf{A}$; The abundance matrices $\mathbf{S}_i (i = 1, 2)$.

```
Random initialization the abundance matrix learning block;
Cascading X1 and X2 into X̄ in spatial dimension;
According to the signal-to-noise ratio (SNR), SVD or PCA method is selected to reduce the dimension of data. The result of dimension reduction is represented as X̂;
while j < P do
    Calculating the projected value of all data on w: Y_j = abs(w X̂);
    Extracting the pixel x_{i_max} which corresponding to the extreme point i_max = arg max(Y_j);
    Updating the shared endmember matrix A = [A, x_{i_max}], j = j + 1;
    Recalculating the vector w which orthogonal to every vector of the endmember matrix: w = ((I − AA#)w)/(‖(I − AA#)w‖);
end
return the shared endmember matrix A.
while epoch < epoch_max do
    Renew S_i (i = 1, 2) by Eq. (2);
    Calculating the loss function L1 by Eq. (3) and (4);
    epoch = epoch + 1;
end
return the abundance matrices S_i (i = 1, 2).
```

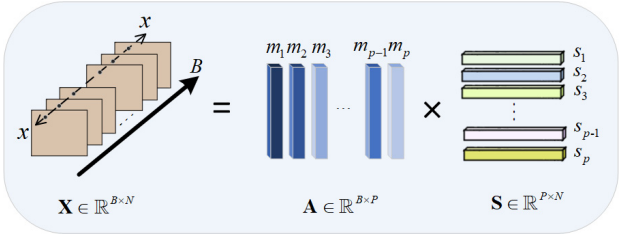


Fig. 4. Illustration of the linear mixture model.

the L_1 norm is adopted to the loss function $\mathcal{L}_1$, which is used to measure the optimization degree of the abundance matrix

$$\mathcal{L}_1 = \sum_{i=1}^{2} \left\| \mathbf{X}_i - \widetilde{\mathbf{X}}_i \right\|_1 \tag{3}$$

where

$$\widetilde{\mathbf{X}}_i = \mathbf{A}\mathbf{S}_i, \quad i = 1, 2. \tag{4}$$

The algorithm flow of endmember sharing-based AMLM is described in Algorithm 1.

B. Hierarchical MSchA-Based DIEM

For the change detection task, there is a strong correlation between the bitemporal images. Nowadays, many deep

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learning-based change detection methods have achieved good results. However, most of them adopted a dual branch structure to extract the corresponding features of bitemporal images, respectively, without fully discussing the correlation between the bitemporal images from the nature of change detection. In this article, we skillfully utilize the correlation difference information and realize the information interaction between bitemporal images. We propose a hierarchical MSchA-based DIEM to analyze the correlation between the bitempral images, in which the enhanced difference feature is extracted by constantly comparing the self-correlation and cross correlation between the abundance matrices of the bitemporal images. Specifically, the basic component of DIEM, MSchA, can capture the difference feature maps. While the hierarchical MSchA can constantly emphasize the difference information, so that the two enhanced difference feature maps are generated to realize the change detection task. The whole process of the proposed DIEM can be formulated by

$$[\mathbf{F}_1, \mathbf{F}_2] = f^t_{\mathrm{MSchA}}\left(f^{t-1}_{\mathrm{MSchA}} \cdots \left(f^1_{\mathrm{MSchA}}(\mathbf{S}_{e1}, \mathbf{S}_{e2})\right)\right) \tag{5}$$

where

$$\mathbf{S}_{ei} = f_{\mathrm{shallow}}(\mathbf{S}_i), \quad i = 1, 2 \tag{6}$$

where $\mathbf{F}_1$ and $\mathbf{F}_2$ represent the two enhanced difference feature maps, $f^t_{\mathrm{MSchA}}(\cdot)$ represents the tth MSchA block, $f_{\mathrm{shallow}}(\cdot)$ contains a convolutional layer with kernel size 3×3 and an ReLU activation function, which is aimed to extract the shallow features, and $\mathbf{S}_{ei}, i = 1, 2$ is the extracted shallow features. Then, the final change map can be represented as follows:

$$\mathbf{CD} = f_{cd}(\mathrm{Concat}(\mathbf{F}_1, \mathbf{F}_2)) \tag{7}$$

where $f_{cd}(\cdot)$ is a classifier made up of the fully connected layers.

The proposed MSchA block is described as follows in detail.

1) Architecture of the Proposed MSchA: In the proposed MSchA, we compare the self-correlation matrix (SCM) with the cross-correlation matrix (CCM) to reassign the weights of the changed and unchanged regions between the bitemporal images. Therefore, the changed regions are retained, while the unchanged regions are suppressed. In this way, the easily distinguished difference information can be effectively extracted.

The architecture of MSchA is shown in Fig. 5. This section will be started with the single-head self-cross-hybrid attention. The abundance matrices $\mathbf{S}_1$ and $\mathbf{S}_2$ are cropped into small patches. Then, the shallow features can be extracted by the convolutinal layer. Each patch corresponds to three matrices, $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$. As shown in Fig. 5, $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ can be obtained by a convolutional layer, which can be expressed as follows:

$$\begin{cases} \mathbf{Q}_i = f^q_{si}\left(\mathbf{S}_{ei}; \mathbf{W}^q_i\right) \\ \mathbf{K}_i = f^k_{si}\left(\mathbf{S}_{ei}; \mathbf{W}^k_i\right) \\ \mathbf{V}_i = f^v_{si}\left(\mathbf{S}_{ei}; \mathbf{W}^v_i\right) \end{cases}, \quad i = 1, 2 \tag{8}$$

where $\mathbf{W}^q_i$, $\mathbf{W}^k_i$, and $\mathbf{W}^v_i$ represent trainable parameters in the convolution operations, and $f^q_{si}(\cdot)$, $f^k_{si}(\cdot)$, and $f^v_{si}(\cdot)$ represent the convolutions with kernel size 3×3.

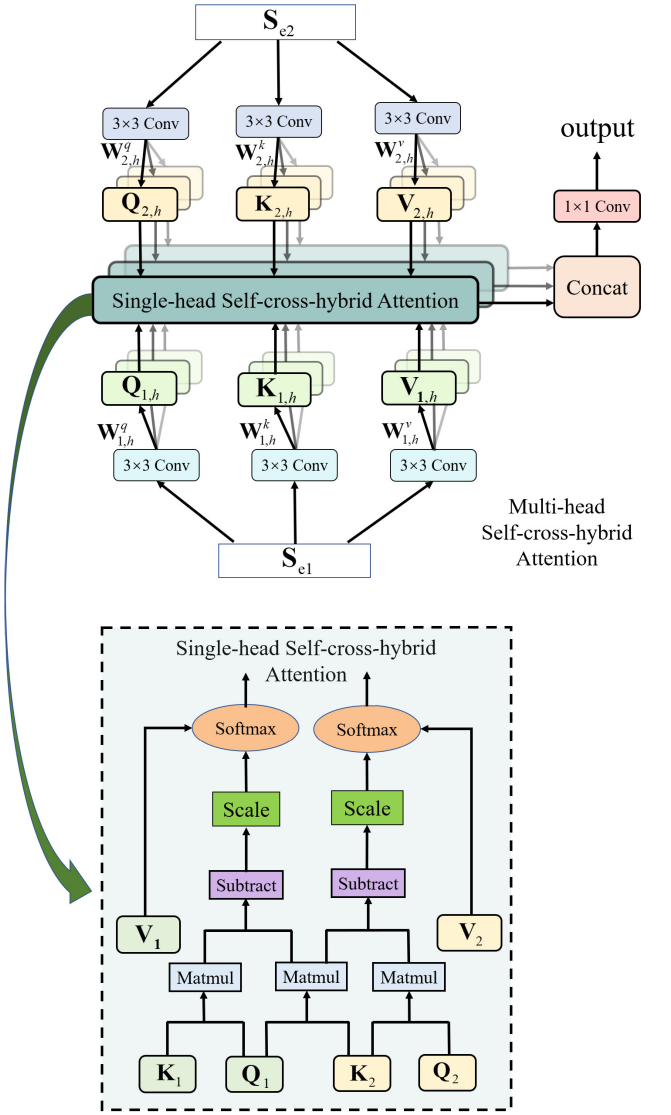


Fig. 5. Architecture of the MSchA.

Afterward, the SCM and CCM of the bitemporal abundance matrices can be calculated by the following formulas:

$$\mathbf{Y}_i = \mathbf{Q}_i \mathbf{K}_i{}^{\mathrm{T}}, \quad i = 1, 2 \tag{9}$$

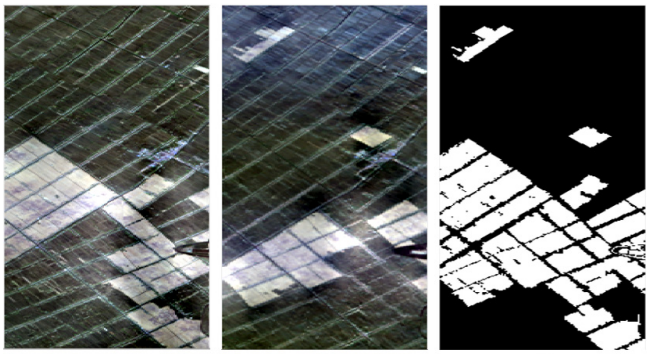
$$\mathbf{Z} = \mathbf{Q}_1 \mathbf{K}_2{}^{\mathrm{T}} \tag{10}$$

where $\mathbf{Y}_i$ represents the SCM, and $\mathbf{Z}$ denotes the CCM. The more similar the bitemporal images are, the stronger the correlation between the abundance matrices is, and the closer the values of $\mathbf{Y}_i$ and $\mathbf{Z}$ are. The difference of $\mathbf{Y}_i$ and $\mathbf{Z}$ is scaled by a particular value d_k to make the gradient more stable. The result is normalized by the softmax function to get the attention weight. Then, the weight matrix multiplied by the values to suppress the unchanged information and obtain the changed information that are easy to distinguish. The single-head self-cross-hybrid attention of bitemporal images is defined as follows:

$$\mathrm{Attention}_1(\mathbf{Q}_1, \mathbf{K}_1, \mathbf{V}_1, \mathbf{K}_2) = \sigma_1\left(\frac{|\mathbf{Y}_1 - \mathbf{Z}|}{\sqrt{d_k}}\right)\mathbf{V}_1 \tag{11}$$

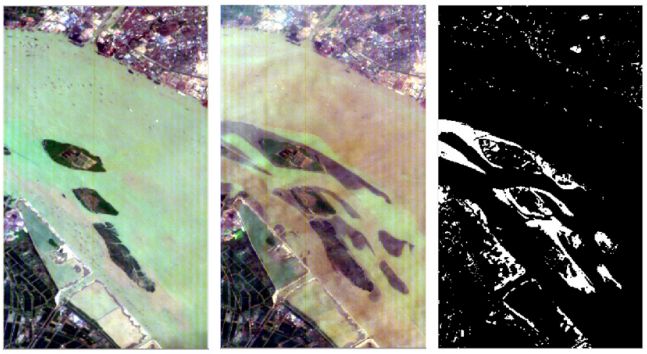
$$\mathrm{Attention}_2(\mathbf{Q}_2, \mathbf{K}_2, \mathbf{V}_2, \mathbf{Q}_1) = \sigma_2\left(\frac{|\mathbf{Y}_2 - \mathbf{Z}|}{\sqrt{d_k}}\right)\mathbf{V}_2 \tag{12}$$

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(a) (b) (c)

Fig. 6. Farmland dataset. (a) Image acquired on May 3, 2006. (b) Image acquired on April 23, 2007. (c) Ground-truth map.



(a) (b) (c)

Fig. 7. River dataset. (a) Image acquired on May 3, 2013. (b) Image acquired on December 31, 2013. (c) Ground-truth map.

where $\sigma_i (i = 1, 2)$ represents the softmax activation function, and the d_k is the dimension of keys.

In order to develop the feature representation ability, we extract features from multiple dimensions to optimize the performance of the module. As shown in Fig. 5, the queries, keys, and values are projected into different subspaces to realize MSchA

$$\begin{cases} \mathbf{Q}_{i,h} = f^q_{mi,h}\left(\mathbf{S}_{ei}; \mathbf{W}^q_{i,h}\right) \\ \mathbf{K}_{i,h} = f^k_{mi,h}\left(\mathbf{S}_{ei}; \mathbf{W}^k_{i,h}\right), \quad i = 1, 2 \\ \mathbf{V}_{i,h} = f^v_{mi,h}\left(\mathbf{S}_{ei}; \mathbf{W}^v_{i,h}\right) \end{cases} \tag{13}$$

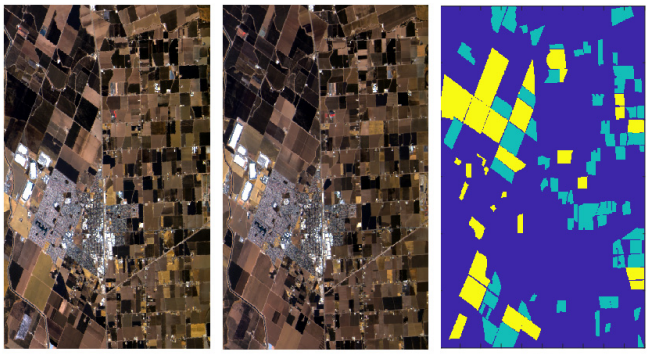
where $h \in [1, \ldots, H]$ denotes the hth head, H represents the number of projection spaces, $\mathbf{W}^q_{i,h}$, $\mathbf{W}^k_{i,h}$, and $\mathbf{W}^v_{i,h}$ are the projection matrices, and $f^q_{mi,h}(\cdot)$, $f^k_{mi,h}(\cdot)$, and $f^v_{mi,h}(\cdot)$ represent the convolutions with kernel size 3×3. The SCM and CCM in hth head of MSchA can be calculated by

$$\mathbf{Y}_{i,h} = \mathbf{Q}_{i,h}{\mathbf{K}_{i,h}}^{\mathrm{T}}, \quad i = 1, 2 \tag{14}$$

$$\mathbf{Z}_h = \mathbf{Q}_{1,h}{\mathbf{K}_{2,h}}^{\mathrm{T}} \tag{15}$$

where $\mathbf{Y}_{i,h}$ and $\mathbf{Z}_h$ denote the SCM and CCM in hth head of MSchA, respectively.

The difference feature maps are the final outputs of the MSchA block, which are the concatenation of the calculating results of each single head attention. The implementation



(a) (b) (c)

Fig. 8. BayArea dataset. (a) Image acquired in 2013. (b) Image acquired in 2015. (c) Ground-truth map.

process corresponding to $\mathbf{X}_1$ is shown as follows:

$$\begin{aligned} \mathbf{M}_1 &= \mathrm{MultiHead}_1(\mathbf{Q}_1, \mathbf{K}_1, \mathbf{V}_1, \mathbf{K}_2) \\ &= \mathrm{Concat}(\mathrm{head}_{1,1}, \ldots, \mathrm{head}_{1,h}, \ldots, \mathrm{head}_{1,H})\mathbf{W}^O_1 \end{aligned} \tag{16}$$

where

$$\begin{aligned} \mathrm{head}_{1,h} &= \mathrm{Attention}_1\left(\mathbf{Q}_{1,h}, \mathbf{K}_{1,h}, \mathbf{V}_{1,h}, \mathbf{K}_{2,h}\right) \\ &= \sigma_{1,h}\left(\frac{\left|\mathbf{Y}_{1,h} - \mathbf{Z}_h\right|}{\sqrt{D_k}}\right)\mathbf{V}_{1,h} \end{aligned} \tag{17}$$

where $\mathbf{M}_1$ represents the result of MSchA that corresponding to $\mathbf{X}_1$, $\sigma_{1,h}$ represents the softmax activation function, the parameter D_k is a specific value selected to keep the gradient stable, and $\mathbf{W}^O_1$ is a projection matrix. The projection is implemented by a convolutional layer with 1×1 kernel size.

Similarly, the implementation process of MSchA corresponding to $\mathbf{X}_2$ is shown as follows:

$$\begin{aligned} \mathbf{M}_2 &= \mathrm{MultiHead}_2(\mathbf{Q}_2, \mathbf{K}_2, \mathbf{V}_2, \mathbf{Q}_1) \\ &= \mathrm{Concat}(\mathrm{head}_{2,1}, \ldots, \mathrm{head}_{1,h}, \ldots, \mathrm{head}_{2,H})\mathbf{W}^O_2 \end{aligned} \tag{18}$$

where

$$\begin{aligned} \mathrm{head}_{2,h} &= \mathrm{Attention}_2\left(\mathbf{Q}_{2,h}, \mathbf{K}_{2,h}, \mathbf{V}_{2,h}, \mathbf{Q}_{1,h}\right) \\ &= \sigma_{2,h}\left(\frac{\left|\mathbf{Y}_{2,h} - \mathbf{Z}_h\right|}{\sqrt{D_k}}\right)\mathbf{V}_{2,h}. \end{aligned} \tag{19}$$

2) Loss Function: The two weighted enhanced difference feature maps $\mathbf{F}_1$ and $\mathbf{F}_2$ are cascaded and fed into the fully connected layers. In addition, the softmax activation function constrains the classification to a probability space. The result could recognize whether the subpixel changed or not. The loss function aims to measure the difference between the predicted value and the ground-truth label, so that the model can be further optimized to make the predicted value as close as possible to the label value. In this article, we applied the cross-entropy loss function, which is performed as follows:

$$\mathcal{L}_2 = -\frac{1}{n}\sum_{k=1}^{n}\left(y_k \log \tilde{y}_k + (1 - y_k)\log(1 - \tilde{y}_k)\right) \tag{20}$$

where n represents the number of samples, y_k denotes the label value in ground truth, and $\tilde{y}_k$ is the predicted value of the proposed method.

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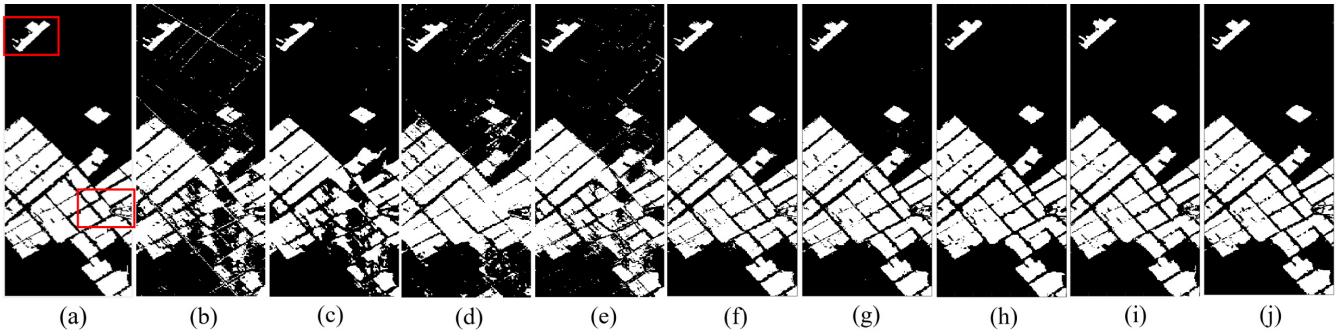


Fig. 9. Visualization results of different methods on the Farmland dataset. (a) Ground-truth map. (b) CVA. (c) PCA. (d) IR-MAD. (e) SVM. (f) ReCNN. (g) BCNNs. (h) BIT. (i) DMPs^{2}raN. (j) AMCAN-HMSchA.

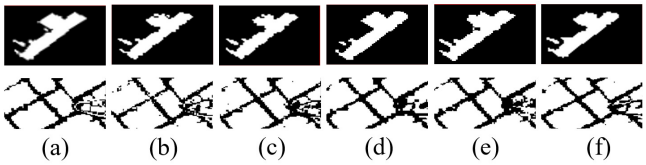


Fig. 10. Enlarged regions (shown in the red frame of Fig. 9) of the deep learning-based methods on the Farmland dataset. (a) Ground-truth map. (b) ReCNN. (c) BCNNs. (d) BIT. (e) DMPs^{2}raN. (f) AMCAN-HMSchA.

III. Experiment Results and Analysis

A. Datasets

To demonstrate the performance of the proposed method, we conduct comparative experiments on three public datasets. Each dataset has three images, which contains a ground-truth map and two HSIs taken at the same area but in different times. The first two are fully labeled datasets captured by Earth Observing-1 (EO-1) Hyperion [38], respectively, Farmland and River datasets. The ground-truth maps are binary images. The black area represents unchanged pixels, while the white area represents changed pixels. The other is Bay Area, which is an incomplete labeled dataset obtained by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor. There are some unmarked areas in the images. The ground-truth map is a three-color map, where the cyan indicates changed, the yellow indicates unchanged, and the violet means undetermined.

1) Farmland Dataset: Farmland dataset[1] is obtained by EO-1. As shown in Fig. 6(a) and (b), the dataset depicts the changes occurred in the farmland near the city of Yancheng, Jiangsu Province, China. These two images are acquired on May 3, 2006 and April 23, 2007, respectively, which have the same size of 420 × 140 pixels. In addition, both images of the dataset contain 242 spectral bands with the spectral range of 0.4–2.5 μm and the spatial resolution of 30 m. After denoising and other preprocessing, the remaining 154 bands can be used for the change detection experiments. The actual changes occurred in the images are shown in the ground-truth map; see Fig. 6(c).

2) River Dataset: As shown in Fig. 7(a) and (b), the River dataset[2] contains two HSIs taken on May 3, 2013 and December 31, 2013, respectively, from a river region in Jiangsu Province, China. The images have a size of 463 × 241 pixels, with a total of 242 spectral bands, and 198 bands are available after noise removal. In addition, the spectral range covers 0.4–2.5 μm, with a spectral resolution of 10 nm and a spatial resolution of 30 m. The real changes of the river region are labeled in the ground-truth image, as shown in Fig. 7(c).

3) Bay Area Dataset: The BayArea dataset[3] has relatively complex ground information, including urban areas and the nearby farmland. As presented in Fig. 8(a) and (b), two images are captured by the AVIRIS sensor in 2013 and 2015, respectively, which covers the city of Patterson (CA), USA. It shows the changes that have taken place on the surface of the city during the past two years. The two HSIs with the same size of 600 × 500 pixels contain 224 spectral bands and cover the spectral range of 0.4–2.5 μm. The spatial resolution is 20 m. The reference ground-truth is shown in Fig. 8(c).

TABLE I
Confusion Matrix

Confusion Matrix		True	
		Changed	Unchanged
Predicted	Changed	TP	FP
	Unchanged	FN	TN

B. Evaluation Metrics

In this article, the four most commonly used evaluation metrics, overall accuracy (OA), kappa coefficient (KC), overall error (OE), and $F1$ score, are adopted to obtain the quantitative evaluation of the experiments. In the calculation, there are four indexes: true positive (TP), true negative (TN), false positive (FP), and false negative (FN). The confusion matrix is shown in Table I. OA represents that the accuracy of the detection results can be calculated as the following:

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{FP} + \mathrm{TN} + \mathrm{FN}}. \tag{21}$$

KC is a measurement criteria of agreement between predicted and reference values. It is a more suitable metric when

[1]https://rslab.ut.ac.ir/data
[2]https://pan.baidu.com/s/14ht8k5H-8ObzHJS6msYZjQ
[3]https://citius.usc.es/investigacion/datasets/hyperspectral-change-detection-dataset

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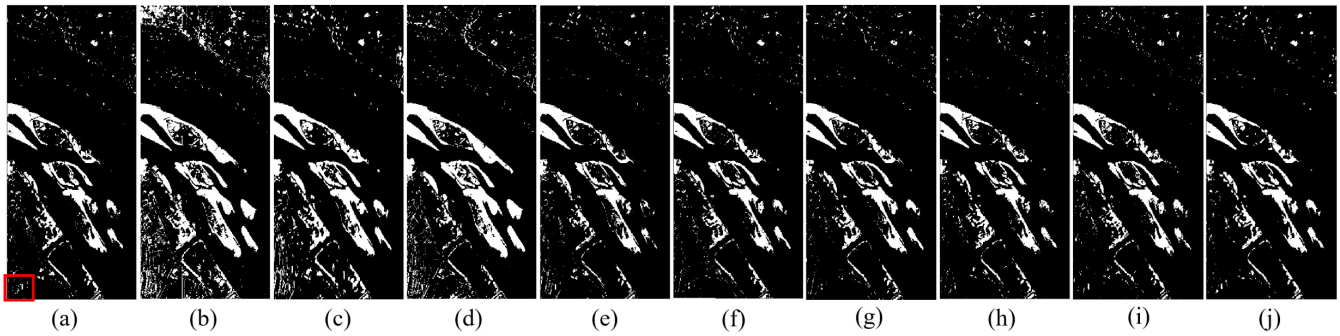


Fig. 11. Visualization results of different methods on the River dataset. (a) Ground-truth map. (b) CVA. (c) PCA. (d) IR-MAD. (e) SVM. (f) ReCNN. (g) BCNNs. (h) BIT. (i) DMPs^{2}raN. (j) AMCAN-HMSchA.

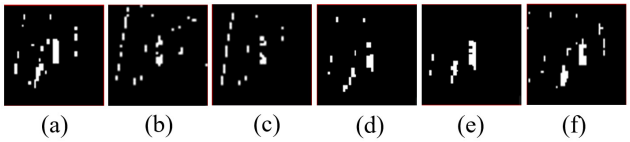


Fig. 12. Enlarged regions (shown in the red frame of Fig. 13) of the deep learning-based methods on the River dataset. (a) Ground-truth map. (b) ReCNN. (c) BCNNs. (d) BIT. (e) DMPs^{2}raN. (f) AMCAN-HMSchA.

the quantities of changed and unchanged samples are quite different. KC can be calculated as follows:

$$\mathrm{KC} = \frac{\mathrm{OA} - \mathrm{PRE}}{1 - \mathrm{PRE}} \tag{22}$$

where

$$\mathrm{PRE} = \frac{(\mathrm{TP}+\mathrm{FP})(\mathrm{TP}+\mathrm{FN}) + (\mathrm{FN}+\mathrm{TN})(\mathrm{FP}+\mathrm{TN})}{(\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN})^2}. \tag{23}$$

OE represents the number of samples, which are misclassified. It can be calculated by the following formula:

$$\mathrm{OE} = \mathrm{FP} + \mathrm{FN}. \tag{24}$$

$F1$ score is the harmonic mean of recall (Re) and precision (Pr) of the network, which is used to evaluate the overall performance of the model. It can be calculated by

$$\text{F1-score} = 2 \cdot \frac{\mathrm{Pr} \cdot \mathrm{Re}}{\mathrm{Pr} + \mathrm{Re}} \tag{25}$$

where

$$\mathrm{Pr} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \tag{26}$$

$$\mathrm{Re} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}. \tag{27}$$

In summary, the value of OE is negatively correlated with the detection accuracy, while $F1$ score as well as OA and KC are positively correlated with the detection performance of the model.

C. Competing Methods and Experimental Settings

To verify the superiority of the proposed method, eight representative methods are selected for comparative experiments. These methods include CVA [11], PCA [13], IR-MAD [14], SVM [39], ReCNN [28], BCNNs [29], DMPs^{2}raN [31], and BIT [33].

The proposed method is conducted on the PyTorch framework with NVIDIA GTX 3090 GPU. Adam optimizer is chosen to train the two modules. In the AMLM training process, the number of training epochs is set to 5000, and the learning rate is set to 0.001. In the training process of DIEM, the batch size is set to 16, and the learning rate is set to 0.0001. We train the module for 200 epochs. The learning rate decays by a factor of 10 after 150 epochs of training. We randomly select 20% of the samples as the training sets for each dataset.

Moreover, since the adjacent pixels in HSI usually correspond to the same ground object, the information of local regions is generally considered to be similar. In this article, two abundance matrices of the bitemporal images are cropped into small patches with the size of 5×5 as the input to extract the difference features. Small patches can ensure the full utilization of local spatial information.

D. Experimental Results

The visualization results generated by different competing methods on three widely used datasets are shown in Figs. 9, 11, and 13. Specifically, as shown in Fig. 9, the detection results of the four traditional methods on the Farmland dataset lost a lot of main changed information. As shown in Fig. 10, the detection performance of the four deep learning-based competing methods has been greatly improved, but the boundary of the detection results is lightly rough. The change maps of different methods on the River dataset are shown in Fig. 11. In the results of the first three methods, there is a lot of noise generated from the false detection. The result of the SVM method has largely reduced FP index, but there are still a large number of changed areas that are not detected. The great detection performance of deep learning-based methods is obvious. However, as shown in Fig. 12, they detect some tiny isolated changed pixels and ignore some details. In addition, it can be seen from Fig. 13 that the shape of changed regions on the BayArea dataset is relatively regular, but the changed regions are fragmented due to the complexity of the ground covered objects. There are many misclassifications in the visualization results of CVA, PCA, and IR-MAD methods, which lead to a large number of noisy spots in the final change

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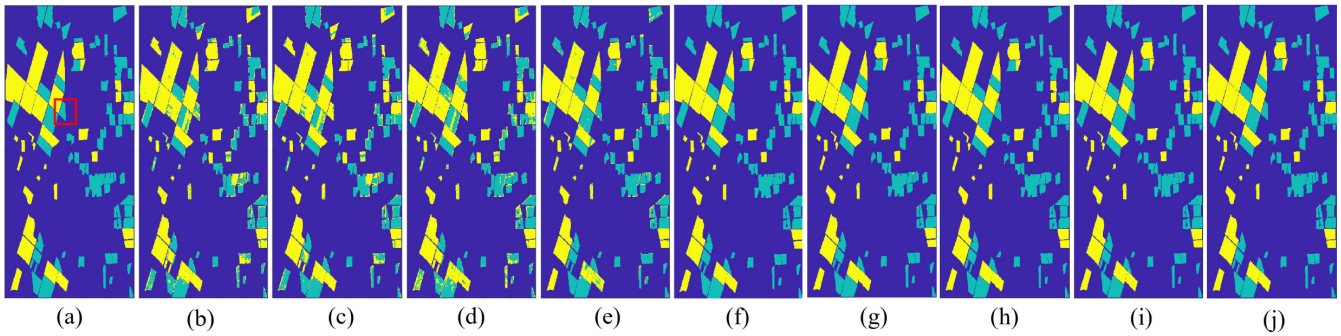


Fig. 13. Visualization results of different methods on the BayArea dataset. (a) Ground-truth map. (b) CVA. (c) PCA. (d) IR-MAD. (e) SVM. (f) ReCNN. (g) BCNNs. (h) BIT. (i) DMPs^{2}raN. (j) AMCAN-HMSchA.

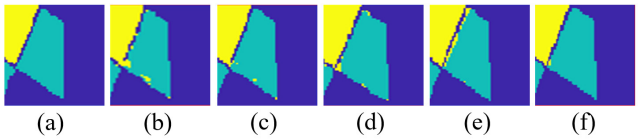


Fig. 14. Enlarged regions (shown in the red frame of Fig. 13) of the deep learning-based methods on the BayArea dataset. (a) Ground-truth map. (b) ReCNN. (c) BCNNs. (d) BIT. (e) DMPs^{2}raN. (f) AMCAN-HMSchA.

map. The detection effect of SVM method has been greatly improved, but it is still slightly inferior to the method based on deep learning. The deep learning-based methods are very effective, and the specific details of the visualization result are not easily distinguished by the naked eye. It can be seen from Fig. 14 that the competing methods ignore some tiny information, leading to some false detection on small scales. By contrast, our method has the highest detection accuracy.

The quantitative assessment of experimental results on three widely used datasets is shown in Table II. The first three classical unsupervised methods cannot achieve an ideal detection result due to the loss of useful information. Because the deep learning-based methods can fully excavate the deep features of HSIs, the detection performance has achieved a cliff-like growth compared with the classical methods. On the Farmland dataset, compared with SVM, the KC of ReCNN increased by 19.5%, while the result of BCNNs has been increased by nearly 20%. The experimental result of our proposed method superior to other competing methods achieves an OA of 0.9799, a KC of 0.9533, and an $F1$ score of 0.9679, respectively. On the River dataset, the performance of deep learning-based methods is better than the traditional methods. The OA and KC values are higher than 0.97 and 0.85, respectively. The proposed method achieves the best results in all competing methods. To be specific, AMCAN-HMSchA achieves an OA of 0.9795, a KC of 0.8686, and an $F1$ score of 0.8798, respectively. On the BayArea dataset, all the methods based on deep learning have achieved good detection performance, which make the value of OE as low as 300. Moreover, the OA, KC, and $F1$ score values of the proposed method are as high as 0.9968, 0.9935, and 0.9970, respectively. The value of OE is as low as 237.

In summary, compared with other competing methods, the proposed method shows the highest OA, KC, and $F1$ score and the lowest OE on three widely used datasets. It can be seen from the subjective experimental result that the proposed method can detect the subtle changes in addition to the significant changes. The reason is that the proposed method solves the problem of mixed pixels caused by the low spatial resolution of the hyperspectral sensor and external interference. More subtle changes can be detected at the subpixel level. Moreover, the structure of the proposed DIEM makes full use of the inherent correlation between bitemporal images, which suppresses the unchanged information and highlights the changed information between the bitemporal images. Therefore, the proposed method possesses high detection accuracy.

E. Ablation Study

1) Number of Endmembers: The selection of the number of endmembers affects the accuracy of the subsequent spectral unmixing and change detection. If the number is estimated too little, the accuracy of endmember extraction will be impure and increases the error between the original image and the reconstructed image. On the contrary, the excessive endmember number estimation will lead to endmember redundancy and a large error of the learned abundance matrix. In order to verify the influence of number of endmembers on the experimental results, different endmembers quantities on three datasets are selected for comparative experiments. The line chart of experimental results is shown in Fig. 15. For the Farmland dataset and the River dataset, the number of endmembers is small due to the simple variety of ground objects covered. It can be seen from the line chart that when the number of endmembers is 8, our method has the best performance on these two datasets. While compared with the first two datasets, the BayArea dataset covers a more complex scene and has more classes of ground objects. Experiments show that when the number of endmembers on the BayArea dataset is estimated to be 14, the performance of our method can be fully exploited. Specifically, the OA and KC values are as high as 0.9968 and 0.9935, respectively.

2) Depth of the Hierarchical MSchA: In the proposed method, the hierarchical MSchA is proposed to analyze the

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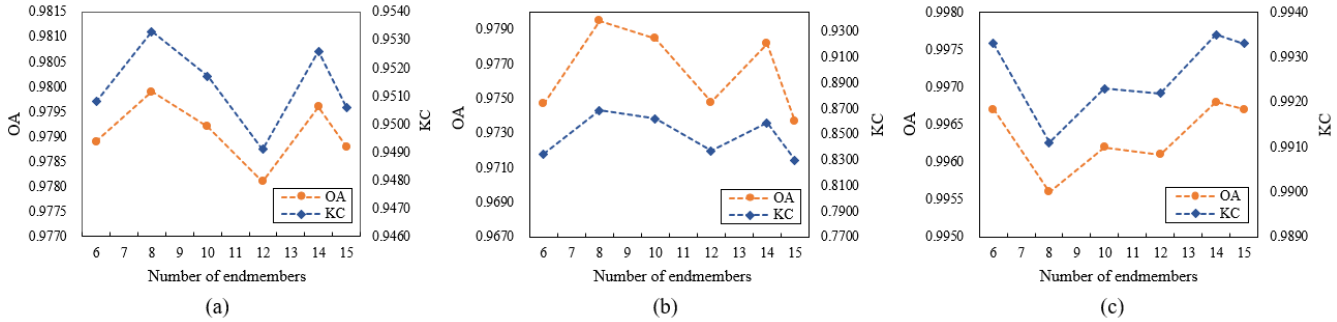


Fig. 15. Detection performance line chart of the proposed method with the different numbers of the shared endmembers on three dataset. (a) Farmland dataset. (b) River dataset. (c) BayArea dataset.

TABLE II

QUANTITATIVE ASSESSMENT OF AMCAN-HMSCHA AND OTHER REPRESENTATIVE CHANGE DETECTION METHODS ON THREE DATASETS

Dataset	Farmland				River				BayArea			
Index	OA	KC	OE	F1-score	OA	KC	OE	F1-score	OA	KC	OE	F1-score
CVA	0.8749	0.6998	7357	0.7882	0.9267	0.6575	8174	0.6955	0.8406	0.6844	11710	0.8433
PCA	0.8892	0.7322	6517	0.8099	0.9538	0.7543	5153	0.7793	0.8433	0.6897	11515	0.8368
IR-MAD	0.8974	0.7632	6034	0.8382	0.9382	0.6798	6887	0.7131	0.8371	0.6782	11969	0.8274
SVM	0.9122	0.7959	5163	0.8598	0.9740	0.8303	2898	0.8444	0.9479	0.8957	3825	0.9504
ReCNN	0.9789	0.9508	1243	0.9662	0.9786	0.8626	2385	0.8743	0.9958	0.9915	312	0.9960
BCNNs	0.9791	0.9515	1228	0.9667	0.9778	0.8572	2482	0.8693	0.9958	0.9917	305	0.9961
BIT	0.9792	0.9515	1224	0.9666	0.9784	0.8571	2410	0.8689	0.9959	0.9917	303	0.9961
DMPs^{2}raN	0.9793	0.9517	1220	0.9668	0.9787	0.8615	2380	0.8732	0.9965	0.9930	257	0.9967
AMCAN-HMSchA	**0.9799**	**0.9533**	**1179**	**0.9679**	**0.9795**	**0.8686**	**2287**	**0.8798**	**0.9968**	**0.9935**	**237**	**0.9970**

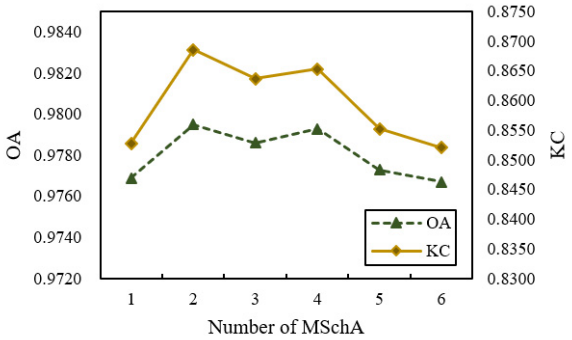


Fig. 16. Detection performance line chart of the proposed method with the different numbers of MSchA.

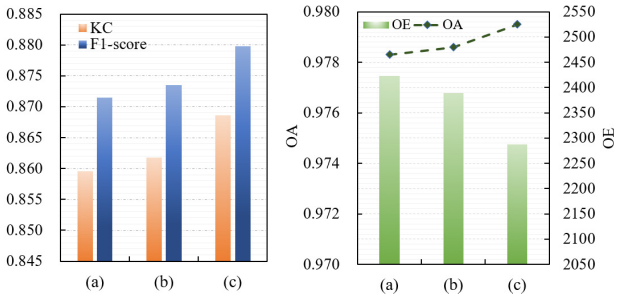


Fig. 17. Detection performance of the variants. (a) Model-SUnSAL. (b) Model-L_1-NMF. (c) AMCAN-HMSchA.

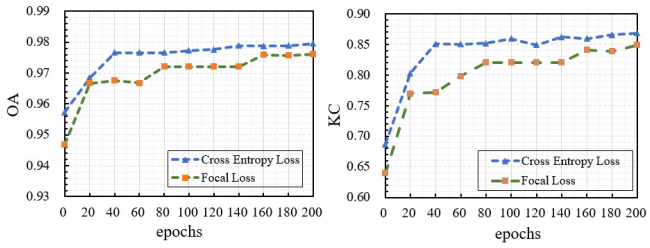


Fig. 18. Detection performance line chart of the proposed method with the different loss functions.

correlation between the input bitemporal HSIs and generate the enhanced difference feature maps. To investigate the influence of depth of the hierarchical MSchA, we conduct experiments on the River dataset with different MSchA layers. The detection performance line chart is shown in Fig. 16. Less stacking layers will lead to insufficient extraction of the difference information. Conversely, if the number of stacked layers is too large, the model structure will be complex, which results in the excessive feature extraction of the training set and the overfitting of the model. Therefore, to achieve the best performance of the network, it is necessary to select an appropriate number of MSchA blocks according to the experiments. It can be seen from the line chart that when the number of MSchA blocks is set to 2, the detection performance of this method is superior to that of other settings. This indicates that the proposed method with two MSchA blocks is sufficient to extract the enhanced difference feature maps on the River dataset.

3) Model Size and Efficiency: The model size and test time of different methods on the River dataset are shown in Table III. Although the traditional methods have high detection

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TABLE III
MODEL SIZE AND EFFICIENCY OF DIFFERENT METHODS

Method	CVA	PCA	IR-MAD	SVM	ReCNN	BCNNs	BIT	DMPs^{2}raN	AMCAN-HMSchA
#Params	-	-	-	-	4.56M	745.22K	3.60M	631.74K	75.82M
Test Time (s)	0.14	0.55	0.90	32.16	16.87	19.45	112.78	185.50	33.40

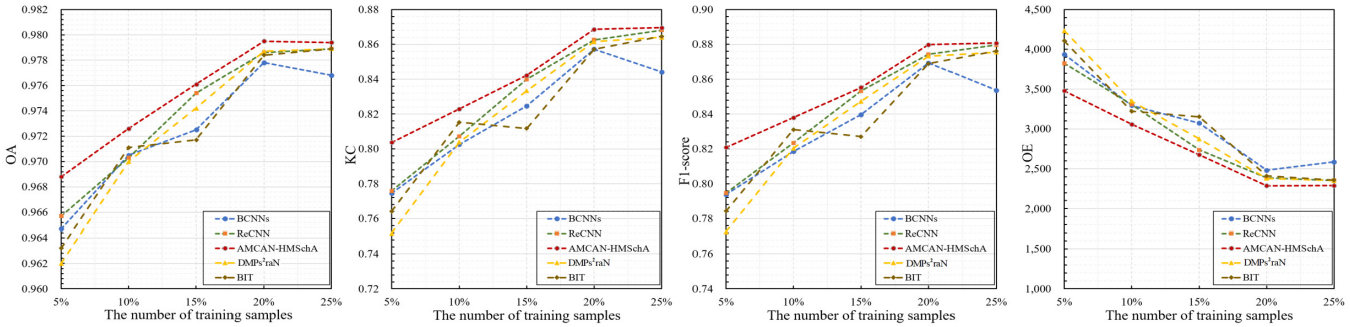


Fig. 19. Detection performance line chart of deep-learning methods with the different numbers of training samples.

efficiency, the detection results of these approaches are not satisfactory. Compared with traditional methods, the deep learning-based methods have larger model size and are more time-consuming. However, the AMCAN-HMSchA achieves the best performance compared with other comparative methods. In future research, we will consider compressing the model size while ensuring the high accuracy of change detection.

4) Effectiveness of AMLM: To verify the effectiveness of the proposed AMLM for learning abundance, we have divided our model structures into three variants, i.e., Model-L_1-NMF using L_1-regularized NMF (L_1-NMF) for unmixing [40], Model-sparse unmixing by variable splitting and augmented Lagrangian (SUnSAL) using variable splitting and augmented Lagrangian for unmixing [41], and our AMCAN-HMSchA with AMLM. We tune the parameters, so that all the variants can obtain their best performance. The experimental results on the River dataset are shown in Fig. 17; although Model-L_1-NMF and Model-SUnSAL can generate satisfactory detection accuracy, the proposed method is still superior to them under the same conditions. Specifically, AMCAN-HMSchA can achieve an OA of 0.9795, a KC of 0.8686, an OE of 2287, and an $F1$ score of 0.8798. Compared with other variants, AMCAN-HMSchA has the highest detection accuracy and the lowest false detection rate. The experimental results confirm the effectiveness of our proposed AMLM.

5) Loss Function: To verify the detection results using cross entropy loss function in this article has sufficient stability. We have replaced the cross entropy loss function with focal loss (FL) [42] for experiments on the River dataset. FL considers not only the problem of imbalance between positive and negative samples, but also the problem of imbalance between easily and difficult distinguishable samples, which can be represented by

$$\mathrm{FL} = -\frac{1}{n}\sum_{k=1}^{n}\left(\alpha\left(1-\tilde{y}_k\right)^{\gamma} y_k \log\left(\tilde{y}_k\right) + (1-\alpha)\tilde{y}_k^{\gamma}\left(1-y_k\right)\log\left(1-\tilde{y}_k\right)\right) \quad (28)$$

where n represents the number of samples, y_k denotes the label value in ground truth, $\tilde{y}_k$ is the predicted value of the proposed method, α is used to balance the number of positive and negative samples, and γ is used to balance the number of difficult and easy samples to classify. In this article, we empirically choose $\alpha = 0.25$ and $\gamma = 2$. As shown in Fig. 18, experimental results with cross entropy loss and FL illustrate that the result using cross entropy loss function has rapid velocity of convergence and sufficient stability.

6) Influence of the Number of Training Samples: We have selected different numbers of training samples to conduct experiments on five deep learning-based change detection methods on the River dataset. The superiority of the proposed method can be proved by the vertical and horizontal comprehensive analyses. It can be seen from Fig. 19 that with the increase in the number of training samples, the performance of the model improves slowly. When the number of training samples is more than 20%, the performance of the model tends to be stable. It can be seen from Fig. 19 that our method performs better than other comparison methods regardless of the proportion of training samples.

IV. CONCLUSION

In this article, we propose a novel method named AMCAN-HMSchA for HSI change detection. AMCAN-HMSchA obtains the difference feature by analyzing the correlation difference information between the bitemporal HSIs in the subpixel level. We design an AMLM based on endmembers sharing named AMLM, which not only can solve the issue of the mixed pixels, but also maps the difference between images to the corresponding abundance matrices. Then, the abundance matrices are imported into the proposed DIEM in parallel to extract the enhanced difference feature. MSchA is the basic structure of the proposed DIEM, in which the feature maps are weighted by the correlation difference between bitemporal images to highlight the changed information and suppress the unchanged information. Experiments on three

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real HSI datasets prove that the proposed method generates a competitive result and even outperforms many state-of-the-art methods.

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